

EDUCATION

University of Illinois Urbana Champaign, Champaign, Illinois, United States

Jul 2023 - May 2026

- **B.S in Statistics | Minor: Mathematics**

INTERESTS & SKILLS

- **Interests:** AI Safety and Alignment, Causal Inference / Causal ML, Bayesian Optimization / Bayesian Modeling, Graph Machine Learning, LLM Security, Anomaly Detection, Meta-Reinforcement Learning, Natural Language Processing, Computer Vision (Zero-Shot Defect Detection), Representation Engineering, Agentic AI, Multimodal LLMs (Vision-Language Models)
- **Languages:** Python, R, SQL, JavaScript, TypeScript, HTML, CSS, LaTeX
- **Models, Tools & Frameworks:** PyTorch, Hugging Face Transformers, vLLM, DeepSpeed, PEFT / LoRA, DPO, LangGraph, FastAPI, Next.js, DGL, NebulaGraph, GPyTorch, BoTorch, scikit-learn, DoWhy, EconML, CausalML, PyMC, SciPy, xarray, OpenCV, YOLOv8, CLIP / OpenCLIP, Sentence-Transformers, BERT, Docker, SageMaker, MLflow, pandas, NetworkX, JAGS / rjags, Poetry, W&B

PUBLICATION & POSTER

- **Mathematically Optimizing Doctoral Consortium Group Assignment**, Beixuan Jin, Dylan Wu et al., University of Illinois Urbana-Champaign. Submitted to the Undergraduate Research Symposium 2026

RESEARCH EXPERIENCE

Research Assistant @ City University of Hong Kong; Advisor: Prof. Xiaowei Luo

Jul 2025 - Apr 2026

- **Climate Risk Assessment System**
 - Applied the ISI-MIP ATTRICI framework (Python, xarray, netCDF4, PyMC/SciPy) to generate counterfactual climate datasets, adjusting factual observations into stationary scenarios for aviation emission trade-off risk assessment.
 - Ran the framework's 8-step detrending and quantile-mapping pipeline to scale time-series data, fit variable-specific statistical models, estimate factual and counterfactual distributions, and preserve observational rank structure across transformed scenarios.
 - Built probabilistic risk-analysis and visualization workflows to assess CO₂ vs. non-CO₂ aviation emission tradeoffs, using Python/R dashboards to summarize counterfactual climate shifts and scenario feasibility for research decision support.

Research Assistant @ Shanghai Jiao Tong University; Advisor: Prof. Mi Xiang

Jul 2025 - Apr 2026

- **Aviation Personnel Mental Health Tracking Application**
 - Implemented a session-memory component for an OpenCouch-style mental-health support app using LangGraph, FastAPI, PostgreSQL/pgvector, and CoALA-inspired memory design to support continuity across sessions and durable persistence.
 - Structured memory into 3 CoALA-inspired layers — semantic facts, episodic arcs, and procedural rules — enabling the backend to retrieve user check-in summaries, support preferences, and recurring mental-health context before follow-up responses.
 - Connected memory and safety logic across 5 supported interfaces — CLI, Next.js web chat, LiveKit web voice, Telegram DM gateway, and FastAPI backend API — to support longitudinal tracking of aviation personnel's self-reported mental-health status.

WORK EXPERIENCE

Quantitative Research Intern @ Zhongshan Securities Co., Ltd

Jun 2025 - Aug 2025

- Built a Python (Pandas) ingestion and cleaning layer over 100+ listed companies' H1 2025 filings, standardizing heterogeneous accounting line items into a unified schema supporting multi-dimensional comparison of revenue, profit, and gross margin. • Integrated the Wind API for market data extraction (prices, PE ratios, ETF constituents, cash-flow series) and designed an indexed SQL schema over Wind-classified industry taxonomy, tuning composite indexes to the dominant query patterns of the research desk.
- Developed a Streamlit/Plotly dashboard tracking the top-10 companies by net asset value within each industry, with sector-level toggles supporting both single-sector drilldown and multi-sector horizontal comparison.
- Implemented walk-forward (rolling-origin) cross-validation for a volatility research effort, the correct validation scheme for time-series data where random k-fold splits leak future information and produce optimistically biased estimates.
- Authored an internal research primer on risk budgeting and risk parity, synthesizing Meketa and CAIA white papers, CFA Level III curriculum, and Bruder & Roncalli (2012); derived the equivalence between the marginal and covariance-based views of risk contribution, extended the framework from absolute risk to tracking error via contribution-to-active-variance, and quantified how volatility drag causes returns to leverage to saturate rather than scale linearly.

Fullstack Intern @ PhinD LLC

Jan 2025 - May 2025

- Adapted AWS's open-source sagemaker-graph-fraud-detection solution, running its pandas-based preprocessing pipeline via SageMaker Processing to convert IEEE-CIS tabular transaction data into node features, labels, and relational edge lists for fraud classification.
- Evaluated graph-based fraud classifiers using DGL's heterogeneous graph construction and RGCN training against 2 feature-based baselines (MLP and XGBoost) on SageMaker training infrastructure.
- Deployed the end-to-end AWS workflow comprising 2 Lambda-triggered modules (data preprocessing and graph modeling), custom Docker processing images, S3 data channels, IAM roles, and CloudFormation templates for reproducible training and endpoint testing.

AI Software Developer Intern @ NetEase Youdao Information Technology Co., Ltd.

May 2024 - Sep 2024

- **Multilingual OCR Engine Development:** Built a text recognition system supporting 20 languages (Chinese, English, Japanese, Korean, etc.) with 96% accuracy per image, processing speed of 1.5s per image, and daily volume of 100K+.
- **Image Preprocessing Optimization:** Developed image enhancement algorithms with OpenCV for denoising, correction, and contrast adjustment; applied YOLOv8 for text detection, achieving 94.2% localization accuracy and improving recognition success on blurred images

by 30%.

- Multi-Engine Translation Service: Built a translation service with FastAPI, integrating Google Translate, Baidu, and DeepL APIs; supported 50+ languages with 300ms average response time and 500K+ daily translations, achieving 92% accuracy.
- AI-powered Text Polishing: Developed a text polishing system leveraging OpenAI GPT-3.5, supporting academic and business styles; applied prompt engineering for enhanced results, improving readability by 30% and achieving 89% user satisfaction.

RESEARCH WORK REPRODUCTION

- **MERINA: Improving Generalization for Neural Adaptive Video Streaming via Meta Reinforcement Learning (Meta-Reinforcement Learning)**
 - Reproduced a two-stage meta-RL pipeline for adaptive bitrate (ABR) streaming, executing VAE-based imitation learning (train_im_v4) for environment-context encoding followed by PPO-based policy optimization (train_ppo_v6) with GAE, on FCC and HSDPA bandwidth trace datasets.
 - Compared MERINA's meta-adaptation generalization against rule-based baselines (BOLA, RobustMPC) on out-of-distribution traces (Puffer, Oboe), understanding the trade-off between fast few-shot adaptation via latent-space task inference and zero-shot transfer robustness.
- **BLSP: Long-term, medium spatial resolution annual land surface phenology with a Bayesian hierarchical model (Bayesian Modeling)**
 - Reproduced the BLSP hierarchical Bayesian model by configuring the rjags-to-JAGS Gibbs Sampler MCMC pipeline, fitting 6- and 7-parameter double-logistic curves to sparse Landsat EVI2 time series through the FitBLSP(), GetModel(), and coda posterior-sampling chain to estimate pixel-wise SOS/EOS phenometrics with credible intervals.
 - Evaluated the trade-off between interannual variability fidelity and data-sparse robustness inherent to the hierarchical prior design, and set up the Docker-based reproducible environment, including Dockerfile and docker-compose, to resolve JAGS binary and R package version dependencies across platforms.
- **SecAlign: SecAlign: Defending Against Prompt Injection with Preference Optimization (LLM Security Alignment)**
 - Ran SecAlign training through align.py, train.py, run.py, and scripts/secalign.sh with PyTorch, Transformers, DPO, and LoRA to teach the model to prefer legitimate instructions over injected prompts.
 - Configured preference data construction with prompt-injected inputs, secure outputs, insecure outputs, and SecAlign training scripts to compare preference optimization against StruQ-style defensive SFT.
- **Circuit Breakers: Improving Alignment and Robustness with Circuit Breakers (Representation Engineering)**
 - Trained circuit-breaker adapters by wiring CustomTrainer, compute_loss, and CircuitBreakerDataset with Transformers, PEFT LoRA, PyTorch, and DeepSpeed to alter harmful hidden representations while retaining benign behavior.
 - Mapped the training workflow from retain data and circuit-breaker prompts through tokenization masks, hidden-state loss, and LoRA merge utilities to understand representation intervention versus refusal-only alignment.
- **Safe-RLHF: Safe RLHF: Safe Reinforcement Learning from Human Feedback (LLM Safety Alignment)**
 - Executed the full Safe-RLHF training sequence across SFT, reward model, cost model, and PPO-Lag scripts using DeepSpeed, Transformers, Hugging Face datasets, and W&B to study helpfulness and harmlessness as separate optimization signals.
 - Inspected the framework modules under safe_rlhf.algorithms, scripts, and model-training entrypoints to connect actor model, reward model, cost model, KL control, and Lagrangian safety constraint during policy optimization.
- **Latent Adversarial Training: Latent Adversarial Training Improves Robustness to Persistent Harmful Behaviors in LLMs (AI Safety)**
 - Traced latent adversarial attack and training workflows across latent_at modules and experiment notebooks using PyTorch, Transformers, PEFT, and TransformerLens to understand residual-stream perturbation for jailbreak, backdoor, and unlearning tasks.
 - Adapted the experiment setup across notebooks, task installation scripts, and model-loading paths with Hugging Face, FastChat, DeepSpeed, and lm_eval to handle dependency compatibility and local runtime execution.
- **LogLLM: LogLLM: Log-based Anomaly Detection Using Large Language Models (Anomaly Detection / System Security)**
 - Reproduced the BERT, projector, and Llama classification pipeline on HDFS/BGL-style log sequences, aligning semantic log embeddings with decoder-based sequence classification for parser-free anomaly detection.
 - Configured Python 3.8, CUDA 12.1, Torch 2.4, Hugging Face model paths, and dataset-specific sliding/session windows to resolve environment setup and large-log preprocessing issues..

PERSONAL PROJECTS

- **Price Elasticity Estimation via Double Machine Learning | scikit-learn, pandas, RandomForest, OLS, parquet, Kaggle Online Retail, Causal Inference**
 - Built a DML-based price elasticity pipeline with cross-fitting, engineered product/temporal/stock confounders, residualized log-price and log-quantity using RandomForest models, and estimated debiased causal elasticity via OLS.
- **Introspective Safety Alignment Pipeline | vLLM, DPO, Transformers, FastAPI, OpenAI API, DeepSpeed, Python, Safety Alignment**
 - Built a STAIR's SI-MCTS safety-alignment workflow, tuning MCTS generation, actor prompts, model checkpoints, reward rules, and Python 3.11 dependencies to generate pairwise DPO data and compare CoT-style SFT with step-level DPO safety tuning.
- **Customer Incentive Recommendation System | DoWhy, EconML, causal-learn PC, NetworkX, MLflow, pygraphviz, Databricks CPU, Causal Inference**
 - Customized a Databricks causal ML accelerator spanning causal discovery, effect identification, DML estimation, recommendation, and refutation, using background knowledge and graph corrections to estimate promotion effects from biased historical allocation data.